

LOGIC AND PROOFS

Subtitle

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OUTLINE

1. TRUTH TABLES OF COMPOUND PROPOSITIONS

2. APPLICATIONS OF PROPOSITIONAL LOGIC

Compound Propositions

- Compound propositions can be built using negation, converse, contrapositive, conjunctions, disjunctions, and implications.
- Example $(p \rightarrow q) \vee (q \wedge p) \vee (q \leftrightarrow p)$
- Note that the truth tables for the above has rows 2^2 , since 2 propositional variable are involved.

Precedence of Logical Operators

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Figure: Precedence of Logical Operators

Constructing Truth Tables

Example: Construct a truth table for $(p \vee \neg q) \rightarrow (p \wedge q)$.

Solution The truth table consists of 2 propositional variables, there are 2^2 rows in the table. The truth table is shown below.

The Truth Table of $(p \vee \neg q) \rightarrow (p \wedge q)$.					
p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

Practice: Construct a truth table for $p \vee q \rightarrow \neg r$.

Construct the truth tables for each of the following.

- $(p \oplus q) \rightarrow (p \wedge q).$
- $(p \rightarrow q) \vee (q \wedge p) \vee (q \leftrightarrow p).$
- $(p \leftrightarrow q) \oplus (\neg p \oplus \neg q).$
- $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r).$

Translating Propositional Logic Statements

1. Identify atomic propositions.
2. Determine appropriate logical connectives.

Example: If I go to Riverwalk or to the shops, I won't complete my tutorial.

p - I go to Riverwalk.

q - I go to the shops.

r - I complete my homework.

The translation would be:

$$(p \vee q) \rightarrow \neg r.$$

Practice: Translate the following logical statements.

- You can get a free pie on Thursday if you buy a pie or a drink.
- The automated reply cannot be send when the system is full.

Translating Propositions

1. q : You can ride a bike.
2. r : You are under 1.2m tall.
3. s : You are older than 16 years old.

Example: Translate $(r \vee \neg s) \rightarrow \neg q$.

if r or not s , then not q .

Replacing: If you are under 1.2m tall or younger than 16 years old, then you cannot ride the bike.

Logic Puzzles

Example There was an island with two kinds of inhabitants, knights, who always tell the truth, and their opposites, knaves, who always lie. You encounter two people A and B. What are A and B if A says “B is a knight” and B says “The two of us are opposite types”?

Solution:

Let p : A is a knight, and q : B is a knight.

$$p \wedge q$$

$$p \wedge \neg q$$

$$\neg p \wedge q$$

$$\neg p \wedge \neg q$$

First, consider A is a knight, i.e. p is true. If A is a knight, then he is telling the truth when he says that B is a knight, and q is true, and A and B are the same type.

However, if B is a knight, then B's statement that A and B are of opposite types, i.e. $(p \wedge \neg q) \vee (\neg p \wedge q)$ would have to be true (A and B are both knights). Therefore A is not a knight, i.e. p is false.

If A is a knave, A's statement that B is a knight is a lie, i.e. “ q is false and B is also a knave. If B is a knave, then B's statement that A and B are opposite types is a lie, which is consistent with both A and B being knaves. Therefore, both are knaves.

Logic Circuits

A **logic circuit** (or **digital circuit**) receives input signals $p_1, p_2, \dots, p_n$ each a bit, and produces output signals each a bit $s_1, s_2, \dots, s_n$.

Complicated digital circuits can be constructed from three basic circuits, called gates.

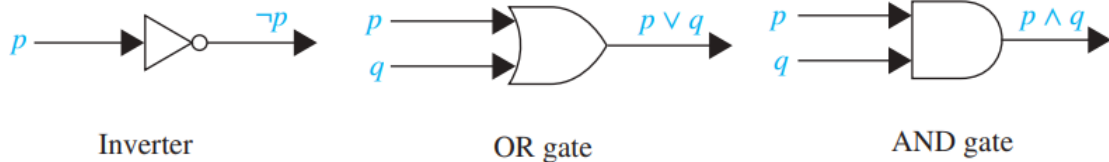
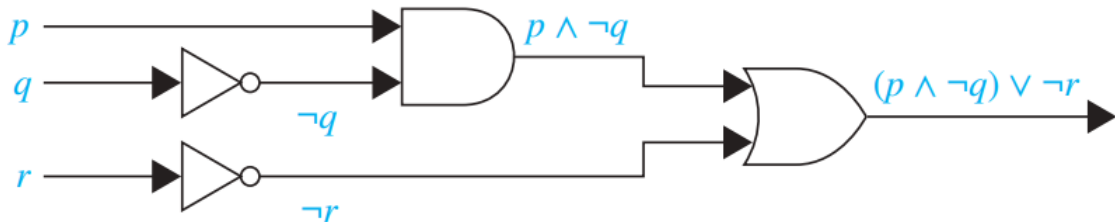


Figure: Basic Logic Gates

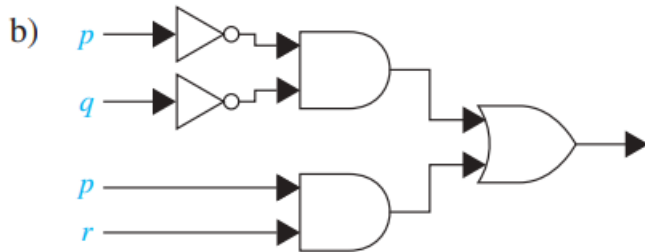
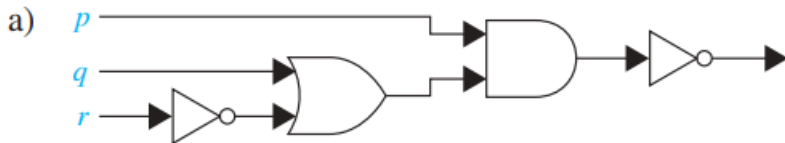
Output of the Combinatorial Circuit



The AND gate takes input of p and $\neg q$, the output of the inverter with input q , and produces $p \wedge \neg q$.

The OR gate takes input $p \wedge \neg q$ and $\neg r$, the output of the inverter with input r , and produces the final output $(p \wedge \neg q) \vee \neg r$.

Find the output of each of these combinatorial circuits.



Construct a combinatorial circuit using inverters, OR gates, and AND gates that produces each of the following outputs from input bits p, q, and r.

1. $(p \wedge \neg r) \vee (\neg q \wedge r).$
2. $((\neg p \vee \neg r) \wedge \neg q) \vee (\neg p \wedge (q \vee r)).$

The End